19	**A**	Using $a = \frac{F}{m}$ and $F = Eq$, $\therefore a = \frac{Eq}{m}$ proton: $a_{proton} = \frac{Eq}{m}$ alpha particle: $a_{alpha} = \frac{E(2q)}{4m} = \frac{a_{proton}}{2}$ Using v = u + at, $v_{proton} = 0 + at$ $v_{alpha} = 0 + \frac{a}{2}t$ Ratio = 0.5
20	**B**	Electric potential decreases along the direction of electric field strength